

Zhengyang Yu

Curriculum Vitae

San Diego & Bay Area

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Education

2025 – Jun 2028 **Bachelor of Science in Computer Science**, *University of California, San Diego*, San Diego, CA

Expected Graduation: June 2028

Relevant Coursework: Advanced Data Structures, Design and Analysis of Algorithms, Computer Organization, Machine Learning, Systems Programming

2023 – 2025 **Associate of Science in Computer Science**, *Laney College*, Oakland, CA

Projects

2026 **Quotaquarium**, *Personal Project*

- Designed and built standalone ESP32-S3 firmware that visualizes Codex quota data as an animated, touch-controlled aquarium with multi-account support, reset countdowns, weather, and configurable alerts.
- Implemented adaptive HTTPS refresh scheduling on dedicated FreeRTOS tasks, RGB565 framebuffer rendering in PSRAM, IMU-driven rotation and water physics, and device-code authorization with bounded persistent logging.

2025 **HTML to ePub Converter**, *Personal Project*

- Engineered a converter to transform HTML content into ePub format, implementing scalable parsing and metadata embedding.
- Enhanced reliability and efficiency through structured error handling and modular component design.

2024 **Home Assistant Presence Plugin**, *Personal Project*

- Built a Home Assistant plugin detecting real-time room occupancy using entry/exit sensor data.
- Designed event-driven architecture integrating Python scripts and RESTful APIs for smart lighting automation.

Skills

Programming C++, Python, Java, C

Embedded ESP32-S3, FreeRTOS, PlatformIO, touch and IMU integration

AI Development Codex-assisted implementation, refactoring, debugging, and technical documentation

Concepts Algorithms, Data Structures, Object-Oriented Design, Software Testing, Debugging

Tools Git, Docker, VS Code, Linux, Fusion 360, Adobe Creative Suite

Activities

2024–2025 Computer Consultant, Oakland Public Library - Elmhurst Branch

Supported community members with computer troubleshooting and taught basic software skills.

2024 Tutor, CTN (Community Technology Network)

Provided one-on-one tutoring to help adults improve digital literacy skills, including computer operation and internet usage.

2024 Member, Laney College MESA Program

Participated in STEM enrichment workshops and peer collaboration.

2025–Present Member, UCSD Envision Makerspace